

CH11.2 平面向量的應用

$$1. -\frac{1}{3} \text{ 或 } 3 \quad 2. \frac{13\sqrt{34}}{34} \quad 3. (E) 4.35.5$$

$$6. 3x - 2y + 1 = 0 \quad 7. 18 \quad 8. \left(-252, \frac{503}{2}\right) \quad 9. 4$$

$$10. 6 \quad 11. -\frac{7}{8} \quad 12. \text{ 見詳解}$$

解析)

$$1. \vec{n}_1 = (2, 1), \vec{n}_2 = (1, a)$$

$$\cos\theta = \pm \frac{\vec{u}_1 \cdot \vec{u}_2}{|\vec{u}_1||\vec{u}_2|}, \frac{1}{\sqrt{2}} = \cos 45^\circ = \pm \frac{2+a}{\sqrt{5} \times \sqrt{1+a^2}}$$

$$\Rightarrow \frac{1}{2} = \frac{(2+a)^2}{5(1+a^2)}$$

$$\therefore 5 + 5a^2 = 2a^2 + 8a + 8$$

$$3a^2 - 8a - 3 = 0, (3a+1)(a-3) = 0$$

$$a = -\frac{1}{3} \text{ 或 } 3$$

$$2. \begin{cases} x = -2 + 5t \\ y = 2 - 3t \end{cases} \Rightarrow \begin{cases} 3x = -6 + 15t \dots\dots \textcircled{1} \\ 5y = 10 - 15t \dots\dots \textcircled{2} \end{cases}$$

$$\textcircled{1} + \textcircled{2} \text{ 得 } 3x + 5y - 4 = 0, \text{ 又 } P(2, -3)$$

$$\therefore d(P, L) = \frac{|3 \times 2 + 5 \times (-3) - 4|}{\sqrt{9+25}} = \frac{13\sqrt{34}}{34}$$

$$3. \text{ 令對稱點坐標為 } (x, y)$$

$$\begin{cases} 3 = \frac{1+x}{2} \\ 2 = \frac{2+y}{2} \end{cases} \therefore \begin{cases} x = 5 \\ y = 2 \end{cases}$$

故選(E)

4. 令 $A(2, -1)$

$$\begin{aligned}\overline{PA} &= \sqrt{(x-2)^2 + (y+1)^2} \geq d(A, L) \\ &= \frac{|3 \times 2 - 4 \times (-1) + 5|}{\sqrt{3^2 + (-4)^2}} = \frac{15}{5} = 3\end{aligned}$$

$\therefore$ 最小值為 3

5. $\overrightarrow{AB}$ 的參數式為 $\begin{cases} x = -47 + 308t \\ y = 11 - 72t \end{cases}, 0 \leq t \leq 1$

$\because$ 308 與 72 之最大公因數為 4

$\therefore t = \frac{0}{4}, \frac{1}{4}, \frac{2}{4}, \frac{3}{4}, \frac{4}{4}$ 時, x 、 y 皆為整數即 $\overline{AB}$ 上有 5 個格子點

6. 令 A 為直線上一點 (x, y)

$$\begin{aligned}\overrightarrow{PA} &= (x-1, y-2) \perp \vec{n} = (3, -2) \\ \Rightarrow 3(x-1) - 2(y-2) &= 0 \Leftrightarrow \underline{3x - 2y + 1 = 0}\end{aligned}$$

$$\begin{aligned}7. \begin{vmatrix} 2a & 2b \\ 3c+e & 3d+f \end{vmatrix} &= \begin{vmatrix} 2a & 2b \\ 3c & 3d \end{vmatrix} + \begin{vmatrix} 2a & 2b \\ e & f \end{vmatrix} \\ &= 2 \times 3 \begin{vmatrix} a & b \\ c & d \end{vmatrix} + 2 \begin{vmatrix} a & b \\ e & f \end{vmatrix} \\ &= 6 \times 2 + 2 \times 3 = 12 + 6 = 18\end{aligned}$$

$$8. \Delta = \begin{vmatrix} 2010 & 2014 \\ 2012 & 2016 \end{vmatrix} < (-1) = \begin{vmatrix} 2010 & 2014 \\ 2 & 2 \end{vmatrix}$$

$$= 4020 - 4028 = -8$$

$$\Delta_x = \begin{vmatrix} 1 & 2014 \\ 0 & 2016 \end{vmatrix} = 2016$$

$$\Delta_y = \begin{vmatrix} 2010 & 1 \\ 2012 & 0 \end{vmatrix} = -2012$$

$$x = \frac{\Delta_x}{\Delta} = \frac{2016}{-8} = -252, y = \frac{\Delta_y}{\Delta} = \frac{-2012}{-8} = \frac{503}{2}$$

$\therefore$ 對數 $(x, y) = \left(-252, \frac{503}{2}\right)$

$$9. \Delta = \begin{vmatrix} 3 & -2a + \frac{2}{2} \\ a-3 & -2 \end{vmatrix} = -6 - (a-3)(-2a+2) \\ = -6 - (-2a^2 + 8a - 6) = 2a^2 - 8a = 2a(a-4)$$

$$\Delta_x = \begin{vmatrix} 12 & -2a+2 \\ 4 & -2 \end{vmatrix} = -24 - 4(-2a+2) \\ = 8a - 32 = 8(a-4)$$

$$\Delta_y = \begin{vmatrix} 3 & 12 \\ a-3 & 4 \end{vmatrix} = 12 - 12(a-3) = -12a + 48 \\ = -12(a-4)$$

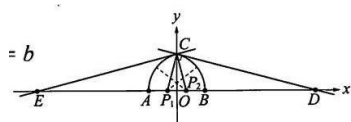
$\therefore$ 有無限多組解

$$\therefore \Delta = 0, \Delta_x = 0, \Delta_y = 0 \therefore a = \underline{4}$$

10. 已知消失點為 O 點，故由單點透視繪圖原理可得到

$$\overline{BC}:\overline{FG} = \overline{OB}:\overline{OF} = \overline{OA}:\overline{OE} = 3:7 \\ \Rightarrow \overline{OA}:\overline{OE} = 3:7 \Rightarrow \overline{OA}:(\overline{OA}+8) = 3:7$$

即 $3\overline{OA} + 24 = 7\overline{OA}$, 故 $\overline{OA} = \underline{6}$



11. 假設 $\angle P_1CP_2 = a$

$$\angle P_2CD = \angle P_1CE = b$$

則 $a + b = 90^\circ$,

且頂角 $\theta = b + 90^\circ$

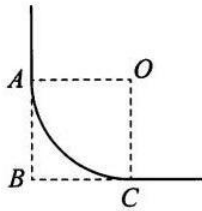
故 $\cos\theta = \cos(b + 90^\circ) = \cos[(90^\circ - a) + 90^\circ] = \cos(180^\circ - a) = -\cos a$

假設 $\overline{AP_1} = \overline{P_1P_2} = \overline{P_2B} = 2$, 則 $\overline{CP_1} = \overline{CP_2} = 4$

在 ΔP_1CP_2 中

$$\cos a = \frac{\overline{CP_1}^2 + \overline{CP_2}^2 - \overline{P_1P_2}^2}{2\overline{CP_1} \times \overline{CP_2}} = \frac{4^2 + 4^2 - 2^2}{2 \times 4 \times 4} = \frac{7}{8}$$

所求 $\cos \theta = -\cos a = -\frac{7}{8}$



12. 圓心角 $\theta = \frac{\pi}{2}$, 半徑 $r = 10$ mm

(1) 弧長 $s = r\theta = 10 \times \frac{\pi}{2} = 5\pi$

$$\approx 15.70796327$$

$$\approx \underline{15.7 \text{ mm}}$$

(2) 所求面積為正方形 $OACB$ 面積 - 扇形 OAC 面積

$$\text{即 } 10^2 - \frac{1}{2} \times 10^2 \times \frac{\pi}{2} = 100 - 25\pi \approx 21.46018366$$

$$\approx \underline{21.5 \text{ mm}^2}$$